



Parental Burnout Across the Globe During the COVID-19 Pandemic

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Abstract: The COVID-19 pandemic has affected all societies worldwide. The heightened levels of stress that accompanied the crisis were also expected to affect parenting in many families. Since it is known that high levels of stress in the parenting domain can lead to a condition that has severe consequences for health and well-being, we examined whether the prevalence of parental burnout in 26 countries (9,923 parents; 75% mothers; mean age 40) increased during COVID-19 compared to few years before the pandemic. In most (but not all) countries, analyses showed a significant increase in the prevalence of parental burnout during the pandemic. The results further revealed that next to governmental measures (e.g., number of days locked down, homeschooling) and factors at the individual and family level (e.g., gender, number of children), parents in less (vs. more) indulgent countries suffered more from parental burnout. The findings suggest that stricter norms regarding their parenting roles and duties in general and during the pandemic in particular might have increased their levels of parental burnout.

Keywords: prevalence, COVID-19, parental burnout, culture, indulgence

Impact and Implications. The results of this unique international study by the International Investigation of Parental Burnout – which includes Western and non-Western countries across the globe – point to the importance of considering parental burnout as a syndrome helping to meet specific Sustainable Development Goals (SDGs). Of all the potentially modifiable influences affecting individuals' healthy lives and well-being across the life course (i.e., SDG 3), positive parenting in the early years has the potential to become a common pathway – by fostering social and emotional skills – to promote a range of healthy outcomes in both children and adults. Acknowledging that parenting can be extremely demanding and exhausting for parents who are confronted with specific individual-, family-, and country-level characteristics may give rise to develop programs how to encourage parents to minimize exhaustion in their parenting role and how to adopt nonviolent ways of disciplining children (SDG 16.2). The various individual and cultural factors as well as COVID-19 factors that have been found related to prevalence rates of parental burnout give indications with factors need to be addressed to promote health and well-being of parents and children (SDG 3) and to diminish or prevent violence against children (SDG 16.2).

Worldwide, approximately three-quarters of a million people each day experience the rewards as well as the challenges of becoming new parents (Bornstein, 2019). These new parents have the important task to raise their children and to contribute to their development and well-being. The important role of parenting for children's well-being is nowadays globally accepted and recognized by the United Nations, in alignment with the 2030 Agenda for Sustainable Development that the member states of the United Nations adopted in 2015. Of the 17 Sustainable Development Goals (SDGs) that have been set, several specifically address the family and recognize the family as the center of social life and well-being (SDG 3) and therefore make parenting and the parenting task an integral part of the sustainable goals (United Nations, 2015).

Although parenting is experienced by many parents as satisfying, it also has been recognized as stressful at times (Abidin, 1990; Deater-Deckard, 2014). When parents lack the resources needed to handle the stressful moments related to parenting, they may develop *parental burnout*, a condition comprised of four main dimensions: emotional exhaustion, contrast with the previous parental self, loss of pleasure in the parental role, and emotional distancing from one's child(ren) (Mikolajczak & Roskam, 2018; Mikolajczak et al., 2019). As shown by Mikolajczak et al. (2020), parental burnout has been recognized as a unique syndrome, empirically distinct from job burnout, depression or common parental stress that may occur when risks (demands) are not compensated by enough resources within the family.

When parents experience extremely high levels of exhaustion and emotional distancing in their parenting task, they can become neglectful or violent toward their children (Hansotte et al., 2020). Mikolajczak et al. (2018a) showed a substantial association between experienced parental burnout and neglectful and violent behavior toward the child(ren). Given that the association between parental burnout and violence did not decrease the significance level when controlling for demographic factors and was not assumed at all typical of families with low socioeconomic status, the syndrome of parental burnout represented a threat to children's well-being even in the most educated families. Therefore, the specific target (SDG 16.2) in the 2030 Agenda for Sustainable Development to end all forms of violence against children gives impetus toward a focus on the syndrome of parental burnout in all countries across the globe.

Prevalence Rates of Parental Burnout

A recent initiative known as the International Investigation of Parental Burnout (IIPB) consortium assessed the prevalence of parental burnout in over 40 countries (Western and non-Western) globally before the pandemic (Roskam et al., 2021). The results of data collected in 2018–2019 showed a worrying number of parents experiencing parental burnout worldwide. Importantly, the prevalence rates of parental burnout varied across countries (Roskam et al., 2021), and it appeared that country-specific cultural factors were found to be associated with parental burnout. For example, differences between countries in prevalence rates of parental burnout were found related to the countries' specific cultural value of *individualism*. Parents in more individualistic countries (e.g., Belgium, Poland, United States) reported higher average levels of parental burnout compared to parents living in more collectivistic countries (e.g., Chile, China, Portugal; Roskam et al., 2021). Although more studies are needed to understand the mechanisms that link individualism (IDV) to parental burnout, the increasing demands created by parenting norms and the intensification of parental investment in the more individualistic Euro-American countries might explain this link (Nelson, 2010; Roskam et al., 2021). Moreover, various factors at both the individual and family level were found related to parents' feelings of burnout by their parenting tasks (for a synthesis, see Mikolajczak et al., 2018b). But what about the role of parental burnout in relation to the COVID-19 pandemic that took over the world in 2020?

Parental Burnout During the COVID-19 Pandemic

The outbreak of COVID-19 in 2020 and the attempts to control it had an enormous impact on people's family lives and on parenting tasks (Weaver & Swank, 2020). The COVID-19 pandemic increased the number of stressors (e.g., additional chores, homeschooling, struggling to occupy children while working from home) and limited the access to usual resources (e.g., no reliance on babysitters, no day care, no grandparents to watch over the children, no leisure activities to breathe out from parenting). Since an imbalance between risks and resources underlies parental burnout (Mikolajczak & Roskam, 2018), the COVID-19 pandemic – with its increased potential for risks and stressors – might lead to a rise in parental burnout.

Since state-imposed lockdown measures could place parents at higher risk for parental burnout symptoms (Griffith, 2020) and given that COVID-19 affected families in countries across all five continents, with lockdown measures differing between countries, we examined whether Hofstede's (2001) cultural values, as indicators of cross-cultural differences, were related to parental burnout levels during the pandemic. Hofstede (2001) identified six statistically different dimensions of culture: (1) IDV versus collectivism (i.e., the extent to which individuals are integrated within groups, the strength of ties with other individuals), (2) power distance (PDI; i.e., the extent to which power is expected and accepted to be distributed unequally by less powerful members of organizations, institutions, or society), (3) uncertainty avoidance (UA; i.e., to which extent members feel comfortable or uncomfortable in ambiguous, new, different from usual or surprising situations), (4) masculinity (MAS) versus femininity (i.e., the distribution of values between gender), (5) long-term versus short-term orientation (i.e., the sense of conservatism or traditionalism in an organization, institution, or society), and (6) indulgence (IND) versus restraint (i.e., the extent to which a society allows the fulfillment of basic human desires that are associated with joy in life and happiness). Each country can be positioned on these six dimensions relative to other countries.

By conducting a new data collection wave in the countries participating in the IIPB consortium in 2020 and comparing the 2020 results to findings of data collected in 2018–2019, we sought to investigate whether the prevalence rates and mean levels of parental burnout increased during the lockdown.

The main research questions of this study were:

1. Have prevalence rates and/or mean levels of parental burnout increased in 2020 (Wave 2), measured during the COVID-19 pandemic, compared to 2018–2019 (Wave 1)?

2. Which sociodemographic and family characteristics are associated with higher levels of parental burnout during the COVID-19 pandemic?
3. Which individual and state-imposed governmental lockdown measures during the COVID-19 pandemic have had an effect on the levels of parental burnout?
4. What country-specific cultural values have had an effect on levels of parental burnout during the COVID-19 pandemic?

Given the unique situation of the COVID-19 pandemic across the globe, we were able to assess what country-specific cultural values were related to the relation between lockdown measures taken by the government and the level of parental burnout in a country.

5. To what extent have country-specific cultural values mitigated the strength of the impact of governmental measures on parental burnout?

Methods

Procedure and Participants

For the data collection, members of the IIPB who participated in the first wave of data collection (see Roskam et al., 2021) were contacted again in April 2020. Not all members were able to collect data during the COVID-19 pandemic, usually due to practical reasons. Finally, members of 26 countries agreed and were able to join the second wave of data collection. Six of the countries were not present in the first wave of the data (i.e., Czech Republic, Egypt, Greece, Israel, Lithuania, and New Zealand). Members of the IIPB collected data by varying recruitment procedures (e.g., social media networks, newspaper advertisement, word of mouth, and door-to-door), and the survey was completed either by paper-and-pencil or online. Parents were eligible to be included in the study if they had at least one child, regardless of their age, still living at home. To avoid bias, this study was introduced as a study designed to better understand parental satisfaction and exhaustion around the world. No incentives were offered to participate. See Table E1, the Electronic Supplemental Material, ESM 1, for the data collection procedure in each country. Respondents who did not complete the Parental Burnout Assessment (PBA; Roskam et al., 2018) were not included in the present research. The final sample included 9,923 parents (74.7% mothers and 25.3% fathers) from the following countries: Belgium (14.6%), Burundi (1.7%), Cameroon (2.2%), Finland (10.7%), France (7.2%), Italy (3.2%), Japan (2.0%), Peru (3.0%), Poland (3.3%), Portugal (5.3%), the Netherlands (4.1%), Turkey (2.5%), the United States (2.7%), Vietnam (1.8%),

Czech Republic (3.5%), Egypt (1.5%), Israel (2.5%), New Zealand (0.8%), China (5.8%), Chile (6.8%), Colombia (2.2%), Lithuania (1.7%), Iran (2.1%), South Korea (2.4%), Greece (3.5%), and Uruguay (2.8%). Ethical approval was obtained by the Ethic Review Board of Tilburg University (EC2018.013R) and/or an institutes' ethics board.

Measures

Parental Burnout

Parental burnout was assessed with the PBA (Roskam et al., 2018), a 23-item questionnaire assessing the four core symptoms of parental burnout: emotional exhaustion (9 items; e.g., "I feel completely run down by my role as a parent"), contrast with previous parental self (6 items; e.g., "I tell myself I'm no longer the parent I used to be"), loss of pleasure in one's parental role (5 items; e.g., "I do not enjoy being with my children"), and emotional distancing from one's children (3 items; e.g., "I am no longer able to show my children that I love them"). All questions were scored using a 7-point Likert scale from 0 to 6 (i.e., *never, a few times a year, once a month or less, a few times a month, once a week, a few times a week, every day*). In our analyses, both the summed score, as well as a prevalence rate of parental burnout will be used as outcome variables. First, the parental burnout score is computed by summing the item scores (ranging from 0 to 138): Higher scores reflect higher parental burnout level. The internal consistency of the scale was assessed by use of Cronbach's α and was indicated to be good to excellent within countries, ranging from .87 to .98 (see Table E2 in ESM 1). To indicate to what extent the construct of parental burnout is measured the same across the different countries (with measurement invariances), we refer to Roskam et al. (2021). Second, prevalence rates of parental burnout were computed to compare the waves of data collection. Cut-off scores for parental burnout in the current study were operationalized based on a preregistered independent study (Brianda et al., 2020) using a multimethod and multi-informant analysis strategy and Roskam et al.'s (2021) research. Indicators of parental burnout, such as self-reports of parents, views of external clinical judges, and a biological measure of chronic stress (the hair cortisol concentration), resulted in a validated cut-off criterion value on PBA equal to or greater than 86 (95% CI = 79.49–93.03; see <https://osf.io/ujfb3> for more details about the analysis strategy). According to Roskam et al. (2021), parents were judged to have parental burnout if their score was equal to or higher than 92 (i.e., if they experience *all* 23 symptoms at least once a week or if they experience at least 16 symptoms daily). This cut-off score of 92 was considered the most stringent cut-off (i.e., the most conservative prevalence

value to avoid overdiagnosis of burnout) and was therefore used in the subsequent analyses in the current study.

Individual Sociodemographic Variables – Level 1

In the present study, the following sociodemographic variables were included: gender (0 = *mother*, 1 = *father*), age, educational level (number of successfully completed school years), paid profession (0 = *yes*, 1 = *no*), type of neighborhood (0 = *disadvantaged*, 1 = *average*, 2 = *prosperous*), and financial situation (0 = *very good*, 1 = *good*, 2 = *sufficient*, 3 = *moderate*, 4 = *poor*).

Family Variables

The following factors were included in the analyses as variables at the family level: number of children, having a child in the age category 0–4 years (0 = *no*, 1 = *yes*), having a child with special needs such as a chronic illness or disability (0 = *no*, 1 = *1 child*, 2 = *more than 1 child*), and family type (0 = *other*, 1 = *single parent*, 2 = *two-parent family*).

Individual Lockdown Variables

Respondents were asked to answer questions related to their personal situation during the COVID-19 pandemic such as working from home (0 = *yes*, 1 = *no*), home-schooling your children (0 = *yes*, 1 = *no*) and number of days locked down, and attention a child requires (0 = *independent including lunch*, 1 = *independent excluding lunch*, 2 = *check every hour*, 3 = *constantly soliciting me*).

Country-Specific State-Imposed Lockdown Measures – Level 2

The Stringency Index of the Oxford COVID-19 Government Response Tracker (OxCGRT; University of Oxford & Blavatnik School of Government, n.d.; Hale et al., 2021) was used as an assessment for the state-imposed lockdown measures per country. The OxCGRT collected daily scores for each country on policy indicators (e.g., school and workplace closure), economic policies (e.g., income support), and health system policies (e.g., testing regimes) with regards to the COVID-19 pandemic. In the current study, one score on the Stringency Index was allocated for each country by calculating the average score on this index during the period of data collection in that specific country (see <https://www.bsg.ox.ac.uk/research/research-projects/coronavirus-government-response-tracker>). A review of this scoring procedure suggested a high degree of accuracy in our data collection. As of December 31, 2020, 84.79% of all the data points had not changed, and since June 1, 2020, 87.45% of the data points had not required a revision (i.e., post hoc alterations to the coding scheme and factual errors). Just 0.41% of the observations were escalated by the reviewers for adjudication (0.25% since June 1, 2020). The scores generated using an IRT model were highly correlated

to a linear index ($r = 0.98$), which reinforces the validity of the approach (see Hale et al. 2021 for information on the actual measures of the index and the methodology of the data collection and aggregation procedure).

Cultural Values

Cultural values for each country were assessed with the six dimensions identified by Hofstede (2001, 2011): IDV, PDI, MAS, UA, long-term orientation (LTO), and IND. For the levels of each country, see <https://www.hofstede-insights.com/product/compare-countries>. Each dimension ranges from 0 to 100, with 50 as a mid-level. The rule of thumb is that if a score is under 50, the culture scores relatively low on that scale and if any score is over 50, the culture scores high on that scale. Burundi and Cameroon do not have scores on any of the six cultural values, and for Israel, no score for IND was available (see Table E3 in ESM 1). Researchers have reported the following Cronbach's α coefficients at the country level for each value dimension: .81 (IDV), .84 (PDI), .76 (MAS), .72 (UA), .57 (LTO), and .79 (IND; see Beugelsdijk et al., 2015; Hofstede & Minkov, 2013). In terms of construct validity, significant correlations were reported by Taras et al. (2012) and Guo et al. (2018) between theoretically relevant external criteria like the Human Development Index (i.e., a statistic composite index of life expectancy, education, and per capita income indicators) and IDV (.56, $p < .05$), PDI (–.58, $p < .05$), MAS (.21, $p < .05$), and LTO (.60, $p < .05$) and between the World Giving Index (i.e., a national measure of prosocial behavior) and UA (–.40, $p < .01$) and IND (.54, $p < .01$). Table 1 shows the descriptive statistics for variables of interest in the current 2020 Wave. We additionally checked the main sociodemographic variables of the two waves. The results revealed that gender ($\chi^2 = 50.75$, $p < .0001$), paid profession ($\chi^2 = 18.81$, $p < .0001$), and neighborhood ($\chi^2 = 7.781$, $p < .0001$) did not significantly differ between the two waves. However, this significant level may be due to the large sample sizes. Odds ratios for these variables (gender, OR = 1.28 [95% CI = 1.21–1.35, $p < .001$]; paid profession, OR = 1.14 [95% CI = 1.08–1.21, $p < .0001$]; neighborhood, OR = 1.08 [95% CI = 1.02–1.14, $p < .05$]) indicated very small differences between the 2018 and 2019 and 2020 waves. Family composition ($\chi^2 = 1.84$, $p = .18$), age ($M1 = 39.20$, $SD = 8.90$ vs. $M2 = 39.99$, $SD = 8.08$, $t = -1.25$, $p = .33$), educational level ($M1 = 14.89$, $SD = 4.34$ vs. $M2 = 15.74$, $SD = 4.57$, $t = 0.15$, $p = .89$), and number of children ($M1 = 1.98$, $SD = 1.19$ vs. $M2 = 1.98$, $SD = 1.12$, $t = 0.99$, $p = .45$) were not significantly different between the two waves.

Statistical Analyses

To answer the first research question, we compared the mean levels of parental burnout and prevalence rates of

Table 1. Description of the individual-level variables (i.e., demographic, parental workload, and individual lockdown)

	<i>N</i>	<i>M</i>	<i>SD</i>	%
Parental burnout	9,923	27.44	28.53	
Age	9,895	39.99	8.08	
Gender: male (female)	9,923			25 (75)
Educational level	9,894	15.74	4.57	
Paid profession: no (yes)	9,914			22 (78)
Financial sit: very good	9,051			16
Financial sit: good	9,051			36
Financial sit: sufficient	9,051			26
Financial sit: moderate	9,051			17
Financial sit: poor	9,051			5
Neighborhood: disadvantaged	9,875			7
Neighborhood: average	9,875			63
Neighborhood: prosperous	9,875			30
Number of children	9,705	1.98	1.12	
Problem child: none	9,699			81
Problem child: one	9,699			16
Problem child: >1	9,699			3
Child younger than 4 years: yes (no)	9,923			44 (56)
Family type: two-parent	9,694			78
Family type: single parent	9,694			11
Family type: other	9,694			11
Days locked down	6,682	42.07	34.72	
Work from home: yes (no)	8,400			75 (25)
Attention child: independent including lunch	9,365			35
Attention child: independent excluding lunch	9,365			20
Attention child: Check every hour	9,365			43
Attention child: Constantly soliciting me	9,365			2
Homeschooling: no (yes)	9,179			29 (71)

parental burnout of Wave 1 (2018–2019) and Wave 2 (2020, during the COVID-19 pandemic) by performing a logistic regression analysis and Mann–Whitney *U* tests. For tests for invariance of measures of burnout, we refer to tests as reported in the study of Wave 1 (Roskam et al., 2021).

For the second to fifth research question, a total of five models were fitted in SPSS 26 by use of hierarchical multilevel regression modeling fitted with restricted maximum likelihood. Hierarchical models were run to select the model that had the best fit to the data (i.e., provided the most parsimonious explanation for the data but that still shows an adequate fit). For this purpose, the Akaike's information criterion and the Schwarz's Bayesian information criterion were used, where lower values indicate better model fit (Akaike, 1969; Stoica & Selen, 2004). As a result, the best fitted model will be reported in the results section, in

comparison to the null model. To increase the interpretability of model parameters, predictor variables were rescaled with categorical variables having a meaningful zero-point. Continuous variables were centered on the grand mean. Model 0 presents the empty model and functioned as a reference null model to estimate intraclass correlation. Model 1 introduced the effects of individual sociodemographical variables, Model 2 introduced family factors, Model 3 introduced the individual lockdown measures, Model 4 introduced the country-specific state-imposed lockdown measures and cultural values of the country, and Model 5 included interaction effects between the country-specific state-imposed lockdown measures and cultural values in each country to test whether country-specific cultural values mitigated the strength of the governmental measures on parental burnout.

Results

Prevalence Rates of Parental Burnout in 2020 Compared to 2018–2019

In total, 26 countries participated in 2020 and collected parental burnout data, with 6 out of 26 countries not having participated in the first wave of the data collection (i.e., Czech Republic, Egypt, Greece, Israel, Lithuania, and New Zealand; see ESM 1 for Figures E1 and E2 with percentages of parental burnout in each country at Wave 1 and Wave 2).

The results of a Mann-Whitney U test showed that the average mean level of parental burnout did not significantly differ between the two waves (Wave 1 mean rank = 17.0, Wave 2 mean rank = 22.0, $U = 133$, $z = -1.387$, $p = .172$). However, the prevalence rates of parental burnout were significantly higher in the second wave, compared to the first wave (Wave 1 mean rank = 15.50, Wave 2 mean rank = 23.50, $U = 104.5$, $z = -.2219$, $p = .03$).

To check whether the mean levels and prevalence rates of parental burnout were related to participation or no participation in the COVID-19 wave, we compared the countries that collected data in Wave 1 only with countries that had gathered data in both waves. A Mann-Whitney U test showed no significant differences in mean levels and prevalence rates of parental burnout between the countries that participated in Wave 1 and not in Wave 2 and the countries that participated in both waves (mean levels of parental burnout $U = 213$, $z = -1.76$, $p = .860$, prevalence rates $U = 119$, $z = -.025$, $p = .980$).

The results of a logistic regression analysis (Table 2) to compare prevalence rates of parental burnout during the first wave in 2018–2019 and the second COVID-19 wave showed a significant increase in parental burnout for the total sample (4.2% vs. 5.8%, respectively). The odds of having parental burnout during the COVID-19 wave were 1.39 times higher than the odds of having parental burnout during the pre-COVID Wave 1 ($p < .001$) with varying odds among countries.

Impact of Various Individual- and Country-Level Characteristics on Parental Burnout

Multilevel analyses were performed to answer the research questions concerning the association between individual-level characteristics, governmental measures, and the levels of parental burnout during the COVID-19 pandemic and to what extent differences in country cultural values were related to the increase in the levels of parental burnout. The null model indicates an intraclass correlation of approximately 9% (76.96/833.32). Based on the fit indices, Model 4 showed the best fit to the data (see Table E4 in ESM 1). This

fourth model included all variables that were added in the first three models (i.e., sociodemographic variables, family variables, lockdown variables) and also variables on level two (i.e., stringency of lockdown and cultural values). The results of this final and best fitted model (Model 4) are described below in more detail (Table 3).

Impact of Individual and Family Characteristics on Parental Burnout

With regard to the sociodemographic variables, significant effects were found for gender, age, and the financial situation of the parents. First, the results indicated that women (mothers) showed a higher level of parental burnout compared to men (fathers; $B = 4.49$, $p < .001$). The results for age indicated that the higher the age of the parents, the lower the level of parental burnout ($B = -.15$, $p = .014$). In addition, the results showed that compared to a poor financial situation, a very good ($B = -10.07$, $p < .001$), good ($B = -6.98$, $p < .001$), sufficient ($B = -4.90$, $p = .012$), and moderate ($B = -3.85$, $p = .046$) financial situation decreased the level of parental burnout. The better the financial situation, the lower the level of parental burnout. The family type (e.g., two-parent, single parent, or other) did not show a significant effect on the level of parental burnout.

A significant effect was found for having a child with special needs, having a child younger than 4 years, and the number of children in the household on levels of parental burnout. This indicated that having two or more children with special needs or suffering from medical, physical, emotional, cognitive, or behavioral problems increased the level of parental burnout compared to having no child ($B = -12.31$, $p < .001$) or one child ($B = -4.83$, $p = .02$) with special needs. In addition, the number of children ($B = 1.82$, $p < .001$) increased the level of parental burnout. For those parents having children all older than 4 years, levels of parental burnout significantly decreased ($B = -2.41$, $p = .02$).

Impact of Individual Lockdown Measures on Parental Burnout

For the individual lockdown variables, significant effects were found for days that families were locked down, whether parents had a task of homeschooling their children, and the amount of attention the child(ren) required from them during the day. First, the more days respondents were locked down, the higher the mean level of parental burnout ($B = .05$, $p < .001$). The need for homeschooling significantly increased the levels of parental burnout ($B = 3.21$, $p < .001$). In addition, compared to

Table 2. Results of logistic regression for countries participating in 2018–2019 and 2020 ($n = 20$)^a

Country	2018–2019		2020		Wave 1 vs. Wave 2
	N	Prevalence burnout	N	Prevalence burnout	Odds ratio
Belgium	1,689	8.1	1,445	7.5	0.92
Burundi	187	5.9	216	25.9	5.60***
Cameroon	208	0.5	217	4.6	10.00*
Chile	431	3.9	670	8.5	2.26**
China	722	1.4	578	1.7	1.25
Colombia	95	1.1	221	6.3	6.36
Egypt	267	2.6	152	2.6	1.00
Finland	1730	4.9	1,058	5.8	1.18
France	1,357	5.5	716	6.7	1.23
Iran	448	1.3	213	4.2	3.25*
Italy	350	0.6	315	1.9	3.38
Japan	500	1.8	200	2.0	1.11
Peru	312	1.3	301	3.3	2.65
Poland	457	7.7	329	6.7	0.86
Portugal	407	2.0	522	4.6	2.40*
The Netherlands	221	2.2	409	2.2	0.97
Turkey	452	0.2	249	1.2	5.50
Uruguay	299	0.3	283	0.4	1.06
United States	406	7.9	263	9.1	1.17
Vietnam	271	0.4	182	1.0	3.00
Total	10,622	4.2	8,323	5.8	1.39***

Note. * $p < .05$; ** $p < .005$; *** $p < .001$. ^aPrevalence rates of countries only participating in 2020: Czech Republic ($n = 350$) 3.4%; Greece ($n = 347$) 3.2%; Israel ($n = 245$) 2.0%; Lithuania ($n = 169$) 8.5%; New Zealand ($n = 83$) 3.5%; South Korea ($n = 237$) 5.9%.

parents who had to check on their children every hour, parents with independent children ($B = -16.88$, $p < .001$), and parents that only needed to be present during lunchtime ($B = -9.08$, $p < .001$) showed lower levels of parental burnout. The more attention a child required, the higher the levels of parental burnout.

Country-Specific Cultural Values and the Strength of the Impact of Governmental Measures on Parental Burnout

With regard to the country-level variables, no significant results were found for the stringency index (i.e., the state-imposed lockdown measures). The analyses only revealed a significant effect for the level of IND. The results of our analysis showed that the higher the level of IND in a country, the lower the level of parental burnout ($B = -.26$, $p = .016$).

A final model did not reveal significant results, showing that country-specific cultural values did not mitigate the strength of the governmental measures on parental burnout. Therefore, we refrain from describing this model in detail.

Discussion

Since the family is regarded as the natural and elementary unit of almost all societies, there are many reasons to focus on the role of families and parenting in meeting the current Sustainable Developmental Goals. An increase in families' well-being and a reduction of family violence is considered to be achieved by supporting parents in positive parenting practices and by diminishing stressors in families (Richardson et al., 2020). In this context, the main goal of the present study was to assess whether a stressor like the COVID-19 pandemic – with its related state-imposed lockdown measures – affected the prevalence of parental burnout (as an indicator of a lack of positive parenting practices and risk factor for increased family violence) worldwide. The results of our analyses showed that the global prevalence of parental burnout significantly increased in many countries during the COVID-19 pandemic. These results confirmed that the pandemic affected family life seriously and dovetails with independent studies conducted in the United States (Russell et al., 2020), Italy (Spinelli et al., 2020), and Germany (Calvano et al., 2021).

Table 3. Multilevel unconditional model and (best-fit) conditional Model 4 predicting parental burnout

	Unconditional model		Model 4	
	Est.	SE	Est.	SE
Intercept	−1.33	1.75	22.27***	3.22
Sociodemographic variables				
Gender (<i>ref father</i>)			4.49***	0.88
Profession (<i>ref no</i>)			0.63	0.95
Age			−0.15*	0.06
Educational level			0.03	0.11
Neighborhood (<i>ref prosperous</i>)				
Disadvantaged			1.57	1.59
Average			−0.13	0.84
Financial situation (<i>ref poor</i>)				
Very good			−10.07***	2.06
Good			−6.98***	1.90
Sufficient			−4.90*	1.95
Moderate			−3.85*	1.92
Family characteristics				
Problem child (<i>ref more than 1 child</i>)				
None			−12.31***	1.92
1 child			−4.83*	2.07
Child younger than 4 years (<i>ref yes</i>)			−2.41*	1.01
No. of children			1.83***	0.40
Family type (<i>ref two-parent</i>)				
Single parent			0.32	1.22
Other			0.30	1.13
Individual lockdown variables				
Work from home (<i>ref no</i>)			−0.91	0.91
Homeschooling (<i>ref no</i>)			3.21***	0.90
Days lockdown			0.05***	0.01
Attention child (<i>ref check every hour</i>)				
Independent 0			−16.88***	1.11
Independent excluding lunch 1			−9.08***	1.05
Country-level variables				
Stringency index			0.07	0.22
Power distance (PDI)			−0.04	0.31
Individualism (IDV)			0.07	0.10
Masculinity (MAS)			−0.02	0.09
Uncertainty avoidance (UAI)			0.04	0.07
Long-term orientation (LTO)			−0.16	0.10
Indulgence (IND)			−0.26*	0.09
Individual variance	756.36***	10.75	663.29***	12.66
Country variance	76.96***	22.72	33.96*	16.43
% explained individual variance			12.30	
% explained country variance			55.87	

Note. *** $p < .001$; ** $p < .01$; * $p < .05$.

Although the odds of experiencing parental burnout were higher in most Western and non-Western countries in 2020 than before the COVID-19 pandemic in 2018–2019, the size of the increase varied considerably across countries. Various individual and family factors were found to be related to increases in rates of parental burnout. Mothers, younger parents, those in poor financial situations, those with more than one child with problems or special needs, those with more children living at home, and those having children younger than 4 years evidenced higher levels of parental burnout during the pandemic. These factors also were found to be significantly related to parental burnout in previous pre-COVID studies focusing on parental burnout (Roskam et al., 2021). Having young children (between the age of 0 and 4 years), more children, and children with special needs certainly increases the attention that children require from parents on a daily basis (Lundberg et al., 1994; Mikolajczak et al., 2018a) and increases the levels of stress associated with parenting. This did not change in 2020. In times of COVID-19 when parents may lose jobs and are confronted with financial uncertain situations, it is not surprising that the amount of stress increases and puts an additional strain on family life and parenting practices. People's daily routines changed, and apart from working from home and parenting, parents were fed up with the task of homeschooling their children. This required extra attention and increased level of stress (Griffith, 2020). The results of our analysis are in line with previous findings and support the assumption that strict lockdown measures (i.e., work remote, stay at home, closing of schools and daycare facilities) increased parental burnout (Joyce, 2022; Skjerdingsstad et al., 2021).

With regard to the cultural values discussed by Hofstede (2011), the seemingly protective factor of collectivism (vs. IDV) in experiencing parental burnout (Roskam et al., 2021) disappeared during the pandemic when many families were in lockdown and social contacts were restricted. During the pandemic, living in a more indulgent country appeared to be a protective cultural value for developing parental burnout. The finding that IND is the main contributing cultural value related to parental burnout across countries during the pandemic seems to contrast with the finding of Roskam et al (2021) which showed no significant effect of IND. This difference is particularly interesting because it shows that when external circumstances change, the protective (or vulnerability) role of cultural values may change too. A value that was not found to be particularly protective in *regular* circumstances became particularly protective in a context of extreme stress on the parents. In response to adversity (e.g. a pandemic and lockdown),

individuals from a culture of IND might feel that they still have control over their participation in life activities, while individuals from a background of cultural restraint may have a sense of helplessness and be less actively involved in taking control over their involvement in enjoyable activities. This finding on the protective role of IND opens perspectives for studies at the individual level too and fits well with existing results showing that parents who take leisure time are actually less vulnerable to parental burnout (Piroux & Mehauden, 2018). Moreover, perhaps less strict norms regarding parenting roles and duties during the pandemic, in particular, in indulgent countries can leave room for individuals to adapt to the extraordinary COVID-19 situation. This may subsequently lead to more freedom for parents how to balance work and home or how to shape the parental role.

Recently, studies have outlined how to prevent or decrease parental burnout (Brianda et al., 2020). High levels of parental burnout in families at risk can be reduced by supporting parents on how to deal with social pressure and perfectionism by enhancing their emotion regulation competencies and stress management techniques. Parents who received a brief and focused intervention not only reported significant decreases in parental burnout symptoms and negative emotions but also decreases in neglect and violence toward their children.

Because of varying rates per country, different sample sizes across countries, variations in geographical areas per country, the over-representation of mothers in almost all countries, and potential unmeasured factors that cannot be ruled out, our results must be interpreted with caution. Nevertheless, the results of this unique study involving 26 Western and non-Western countries around the globe point to the importance of considering parental burnout as a syndrome shaped by various individual and cultural factors as well as natural disasters like pandemics. Acknowledging that parenting can be extremely demanding and exhausting for many parents across the globe, particularly in times of the COVID-19 pandemic, may give rise to the development of specific parenting programs. Those programs should focus on minimizing the exhaustion of parents and should aim to support these individuals to adopt nonviolent ways of parenting their children (SDG 16.2) to promote the health and well-being of their offspring (SDG3).

Electronic Supplemental Material

The electronic supplementary material is available with online version of the article at <https://doi.org/10.1027/2157-3891/a000050>

ESM 1. Data collection procedure in each country (Table E1); mean levels, prevalence rates, and reliability scores of parental burnout for each country in 2020 (Table E2); description of the country level variables (stringency index and cultural values; Table E3); prevalence rates of parental burnout for each country in 2018–2019 (pre-COVID) and in 2020 (Figures E1 and E2); fit indices BIC and AIC (Table E4).

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